Array-Based DOA Estimation Using Digital Signal Processing Techniques

January 1, 2026

Project Members

- Alperen Kahraman (Student ID: 2021401276)
- Abdulhalim Kiraz (Student ID: 2021401219)

Project Members Responsibilities

- **Alperen Kahraman:** Responsible for derivation of the beamforming and MUSIC algorithm, Forward- Backward Spatial Smoothing.
- **Abdulhalim Kiraz:** MATLAB implementation of the ULA steering vectors, noise generation, and simulation of the beam patterns.
- **Joint Work:** Both members collaborated on the problem definition, literature review and the analysis of the simulation results.

1 Problem Statement

In modern communication and radar systems, the ability to accurately locate the source of a signal is a fundamental requirement. Traditional single-sensor systems are often insufficient for distinguishing between multiple signals arriving from different directions, especially in environments with significant noise or interference. Consequently, antenna arrays have become an applicable solution.[1].

The primary problem addressed in this project is the estimation of the Direction of Arrival (DOA) for narrowband sources using a Uniform Linear Array (ULA)[1]. While identifying a single strong signal is relatively straightforward, the challenge increases significantly when multiple sources are located close to each other. In such cases, standard methods are limited by the physical size of the array (the Rayleigh resolution limit)[6], causing separate sources to blur into a single peak.

The motivation for this project is to overcome these physical limitations by implementing and comparing different Digital Signal Processing (DSP) techniques. Specifically, the limitations of the conventional Delay-and-Sum Beamformer [2] are analyzed and the superior method, the Multiple Signal Classification (MUSIC) [4] algorithm is demonstrated. Furthermore, the “coherent signal problem” (often caused by multipath propagation), in which standard subspace algorithms fail, is addressed and Forward-Backward Spatial Smoothing is implemented as a solution [5].

2 Introduction

Array signal processing extends the principles of time-domain Digital Signal Processing into the spatial domain. When several antenna elements are arranged in a specific structure, such as a Uniform Linear Array (ULA), the received signals can be combined in a way that strengthens signals coming from desired directions and reduces signals from other directions. This type of electronic control of the antenna array is known as Digital Beamforming. As explained by Steyskal [2], digital beamforming allows the antenna pattern to be adjusted flexibly, so the system can respond to changing signal conditions without the need for mechanical movement.

The most basic method for this purpose is Conventional Beamforming, also known as the Delay-and-Sum method. According to Van Veen and Buckley [3], this technique works as a spatial filter. It applies phase shifts to the signals received at each antenna element so that they add constructively in a chosen direction. Although this method is robust and simple to compute, its performance strongly depends on the array aperture. As a result, it produces wide beams and high sidelobes, which makes it difficult to separate sources that are close in angle.

To achieve high-resolution Direction of Arrival (DOA) estimation beyond the Rayleigh limit[6], subspace-based methods were developed. A well-known method of these is the MUSIC (Multiple Signal Classification) algorithm. Khan et al. [4] explain that MUSIC uses eigen-decomposition of the covariance matrix to divide the observation space into separate signal and noise subspaces. By searching for steering vectors that are orthogonal to the noise subspace, MUSIC generates sharp peaks at the correct angles of arrival, largely avoiding the beamwidth limitations of conventional beamforming.

In this project, MATLAB is used to simulate a ULA and to implement both Conventional Beamforming and the MUSIC algorithm. A comparative literature review and simulation study are carried out to evaluate their performance under different conditions which includes varying Signal-to-Noise Ratios (SNR), numbers of snapshots, and antenna elements. In addition, the failure of standard MUSIC in the presence of coherent signals is examined and spatial smoothing technique[5] is shown to address this problem.

3 Methodology

To find the direction of DOA, various approaches can be used. In this section, firstly, Direction of Arrival will be explained. Then, Conventional Beamforming and MUSIC methods will be introduced to calculate DOA. Finally, the forward-backward smoothing method for correlated signals will be examined.

The derivation follows the geometric properties of the Uniform Linear Array (ULA).[7]

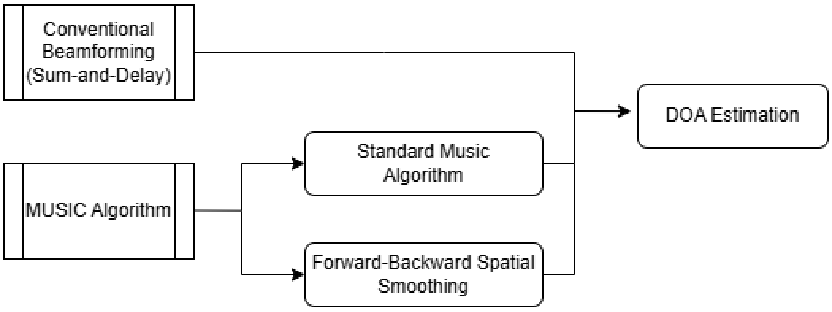

Figure 1: The planning of the methodology

3.1 Theoretical Background of DOA

3.1.1 Geometric Derivation and Signal Model

Consider a uniform linear array where a plane wave impinges on the antennas. Based on the array geometry, the extra distance traveled by the wave to reach the second antenna relative to the first is[8]:

$$\Delta d = d \sin\theta \tag{1}$$

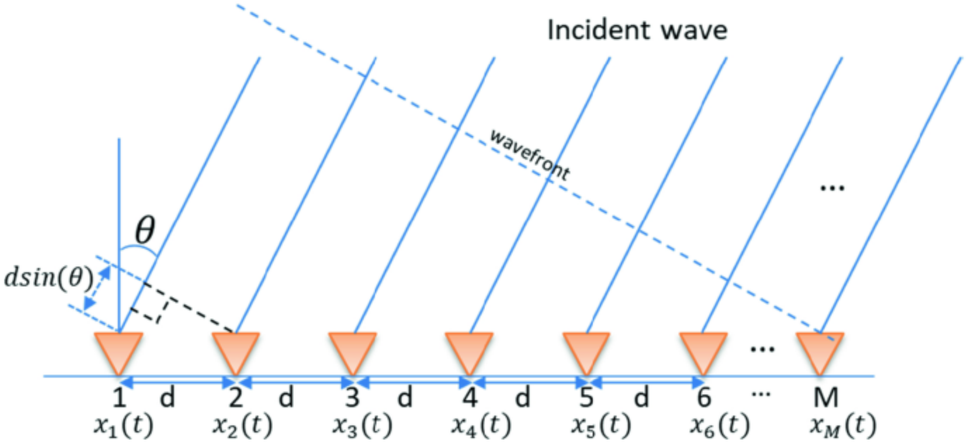

Figure 2: An incident wave striking a 1-D array placed at specific intervals d

Consequently, the time delay τ between the elements is:

$$\tau = \frac{d \sin\theta}{c} \tag{2}$$

where c is the speed of light.

Relating this to the wavelength λ, the fraction of the cycle is the ratio of the extra distance to the total wavelength:

$$\text{Fraction} = \frac{\Delta d}{\lambda} = \frac{d \sin\theta}{\lambda} \tag{3}$$

Since one full wave cycle corresponds to a phase of 2π, the phase shift ϕ between two adjacent antennas is derived as:

$$\phi = -\frac{2\pi}{\lambda} d \sin\theta \tag{4}$$

3.1.2 Steering Vectors and Matrix Formulation

If antenna 1 is designated as the reference (phase 0), antenna 2 is shifted by ϕ, antenna 3 by 2ϕ, and so on. The steering vector $\mathbf{a}(\theta)$, which characterizes the array response to a single source, is defined as:

$$\mathbf{a}(\theta) = \begin{bmatrix} 1 \\ e^{j\phi} \\ e^{j2\phi} \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 \\ e^{-j\frac{2\pi}{\lambda}d\sin\theta} \\ e^{-j\frac{2\pi}{\lambda}2d\sin\theta} \\ \vdots \end{bmatrix} \tag{5}$$

Let $s_1(t)$ be the signal from source 1, $s_2(t)$ from source 2, etc. The voltage measured at the antennas, denoted by vector $\mathbf{x}(t)$, is the sum of the steering vectors multiplied by the respective signals plus noise:

$$\mathbf{x}(t) = \mathbf{a}(\theta_1)s_1(t) + \mathbf{a}(\theta_2)s_2(t) + \cdots + \mathbf{n}(t) \tag{6}$$

This can be written compactly in matrix form:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{n}(t) \tag{7}$$

where $\mathbf{A}$ is the steering matrix containing the steering vectors for all sources.

3.1.3 The Correlation Matrix

To analyze the “pattern of delays” over a large number of measurements, the correlation (covariance) matrix $\mathbf{R}_x$ is computed. For N snapshots, it is estimated as:

$$\mathbf{R}_x = \frac{1}{N}\sum_{t=1}^{N} \mathbf{x}(t)\mathbf{x}^H(t) \tag{8}$$

3.2 Conventional Beamforming

Conventional beamforming (Delay-and-Sum) spatially filters the signal by steering the array to a specific direction. For a target angle θ_{target} corresponding to a phase shift ϕ_{target}, and a scanning angle θ_{scan} corresponding to ϕ_{scan}, the phase shift relation is:

$$\phi = -\frac{2\pi}{\lambda}d\sin\theta \tag{9}$$

A weight vector $\vec{w}$ is defined to align the array phases to the scan direction. To normalize the response, scaling by the number of antennas M is applied:

$$\vec{w} = \frac{\mathbf{a}(\theta_{scan})}{M} \tag{10}$$

The output of the beamformer in the time domain is the inner product of the weights and the received signal:

$$y(t) = \vec{w}^H\vec{x}(t) \tag{11}$$

If the scan angle θ_{scan} matches the true target angle, the phases in $\vec{w}$ perfectly cancel the phases in $\vec{x}$, resulting in coherent addition and maximum output amplitude. If the angles do not match, the addition is destructive, yielding a smaller peak.

3.2.1 Power Calculation

While the beamformer output $y(t)$ represents the instantaneous voltage amplitude, which fluctuates rapidly at the carrier frequency and is susceptible to noise, the primary interest lies in the spatial power distribution. Therefore, we calculate the expected power $P(\theta)$ by averaging the squared magnitude of the output over multiple snapshots. This statistical averaging removes temporal oscillations and suppresses random noise, providing a stable estimate of the signal energy arriving from the scanned direction.

$$P(\theta) = E[|y(t)|^2] = E[y(t)y^H(t)] \tag{12}$$

Calculating $y(t)$ for every snapshot and every angle is computationally expensive. Instead, the power equation can be rewritten using the covariance matrix. Substituting $y = \vec{w}^H\vec{x}$:

$$\begin{aligned} P(\theta) &= E[(\vec{w}^H\vec{x})(\vec{w}^H\vec{x})^H] && (13) \\ &= E[\vec{w}^H\vec{x}\vec{x}^H\vec{w}] && (14) \end{aligned}$$

Since the weights $\vec{w}$ are constant for a given angle, they can be taken out of the expectation:

$$P(\theta) = \vec{w}^H E[\vec{x}\vec{x}^H]\vec{w} \tag{15}$$

Recall that $\mathbf{R}_x = E[\vec{x}\vec{x}^H]$. Thus, the spatial spectrum can be computed efficiently as:

$$P(\theta) = \vec{w}^H\mathbf{R}_x\vec{w} \tag{16}$$

This allows to compute $\mathbf{R}_x$ once and then determine the power for every scan angle using simple matrix multiplication.

3.2.2 Array Resolution Limits

The angular resolution of a Uniform Linear Array (ULA) using conventional beamforming is governed by the null-to-null beamwidth of its main lobe. For a ULA with M elements and inter-element spacing d, the beamwidth is approximately $BW_{NN} \approx \frac{2\lambda}{Md}$.

Choosing $d = \lambda/2$, which corresponds to the spatial Nyquist limit and avoids spatial aliasing, this simplifies to $BW_{NN} \approx \frac{4}{M}$ radians. According to a Rayleigh-type resolution criterion, two sources can be resolved by conventional beamforming only if their angular separation exceeds half the beamwidth, yielding a minimum resolvable separation of $\Delta\theta_{\min} \approx \frac{2}{M}$.

For separations below this limit, conventional beamforming produces a single merged peak, whereas high-resolution methods such as MUSIC may still resolve the sources.

3.2.3 Pseudocode for the algorithm

Algorithm 1 Conventional Beamforming (Delay-and-Sum)

```
Require: Data matrix $X \in \mathbb{C}^{M\times N}$, spacing $d$, wavelength $\lambda$, scan grid $\Theta = \{\theta_i\}$
Ensure: Normalized spatial spectrum $P_{\mathrm{BF,dB}}(\theta)$ over $\Theta$
Estimate covariance: $R_x \leftarrow \frac{1}{N} X X^H$
$k \leftarrow \frac{2\pi}{\lambda}, \quad m \leftarrow [0, 1, \ldots, M-1]^T$
for each scan angle $\theta \in \Theta$ do
    Steering vector: $a(\theta) \leftarrow \exp(-j\,k\,d\,m\,\sin(\theta))$
    Beamformer weights: $w(\theta) \leftarrow \frac{a(\theta)}{M}$
    Power: $P_{\mathrm{BF}}(\theta) \leftarrow \Re\{w^H(\theta) R_x w(\theta)\}$
end for
Normalize: $P_{\mathrm{BF,dB}}(\theta) \leftarrow 10\log_{10}\left(\frac{P_{\mathrm{BF}}(\theta)}{\max_\theta P_{\mathrm{BF}}(\theta)}\right)$ return $P_{\mathrm{BF,dB}}(\theta)$
```

3.3 The MUSIC Algorithm

Unlike beamforming, which scans for power, the Multiple Signal Classification (MUSIC) algorithm estimates DOAs by exploiting the eigen-structure of the covariance matrix.

3.3.1 Covariance Matrix Decomposition

Recalling the signal model $\vec{x} = \mathbf{A}\vec{s} + \vec{n}$, the structure of the covariance matrix $\mathbf{R}_x = E[\vec{x}\vec{x}^H]$ is examined. The terms are expanded as:

$$\begin{aligned}\mathbf{R}_x &= E[(\mathbf{A}\vec{s} + \vec{n})(\mathbf{A}\vec{s} + \vec{n})^H] && (17)\\ &= E[\mathbf{A}\vec{s}\vec{s}^H\mathbf{A}^H + \mathbf{A}\vec{s}\vec{n}^H + \vec{n}\vec{s}^H\mathbf{A}^H + \vec{n}\vec{n}^H] && (18)\end{aligned}$$

Making the standard assumptions that:

1. Signal and noise are uncorrelated ($E[\vec{s}\vec{n}^H] = \mathbf{0}$),
2. Noise is spatially white ($E[\vec{n}\vec{n}^H] = \sigma^2\mathbf{I}$),

The expression simplifies to:

$$\mathbf{R}_x = \mathbf{A}\mathbf{R}_s\mathbf{A}^H + \sigma^2\mathbf{I} \tag{19}$$

where $\mathbf{R}_s = E[\vec{s}\vec{s}^H]$ is the signal correlation matrix.

3.3.2 Eigen-Decomposition

The eigenvectors $\vec{v}$ and eigenvalues λ of $\mathbf{R}_x$ are analyzed:

$$\mathbf{R}_x\vec{v} = \lambda\vec{v} \implies (\mathbf{A}\mathbf{R}_s\mathbf{A}^H + \sigma^2\mathbf{I})\vec{v} = \lambda\vec{v} \tag{20}$$

Rearranging terms:

$$\mathbf{A}\mathbf{R}_s\mathbf{A}^H\vec{v} + \sigma^2\vec{v} = \lambda\vec{v} \tag{21}$$

This equation leads to two distinct cases for the eigenvectors:

Case A: Signal Subspace (E_s)
If $\vec{v}$ lies in the column space of $\mathbf{A}$ (the signal space), then $\mathbf{A}\mathbf{R}_s\mathbf{A}^H\vec{v} \neq 0$. The eigenvalues correspond to the signal power plus the noise power:

$$\lambda_s = \mu_i + \sigma^2 \tag{22}$$

where μ_i are the eigenvalues of the pure signal component without noise.

Case B: Noise Subspace (E_n)
If $\vec{v}$ is orthogonal to the column space of $\mathbf{A}$ (the steering vectors), then:

$$\mathbf{A}^H\vec{v} = \mathbf{0} \implies \mathbf{A}\mathbf{R}_s\mathbf{A}^H\vec{v} = \mathbf{0} \tag{23}$$

Substituting this back into the eigen-equation:

$$\mathbf{0} + \sigma^2\vec{v} = \lambda_n\vec{v} \implies \lambda_n = \sigma^2 \tag{24}$$

Eigenvalues corresponding to signal eigenvectors are $\lambda_s = \mu_i + \sigma^2$, and eigenvalues corresponding to noise eigenvectors are $\lambda_n =\sigma^2$. So the eigenvalues of signals λ_s are the largest eigenvalues.

The remaining eigenvalues are exactly equal to the noise power σ^2.

3.3.3 Subspace Partitioning and Orthogonality

The eigenvalues are sorted in descending order: $\lambda_1 \geq \lambda_2 \cdots \geq \lambda_M$.

- The largest K eigenvalues (where K is the number of targets) correspond to the **Signal Subspace** (E_s).
- The remaining $M - K$ eigenvalues correspond to the **Noise Subspace** (E_n).

The crucial property of MUSIC is that any steering vector $\mathbf{a}(\theta)$ corresponding to a true source direction is orthogonal to the noise subspace. Mathematically:

$$\mathbf{E}_n^H\mathbf{a}(\theta_{true}) = \mathbf{0} \tag{25}$$

This implies that the projection of the true steering vector onto the noise subspace is zero.

3.3.4 The MUSIC Spectrum

To find the DOAs, every possible angle θ is scanned and the squared norm of the projection onto the noise subspace is calculated:

$$d^2(\theta) = \|\mathbf{E}_n^H\mathbf{a}(\theta)\|^2 = \mathbf{a}(\theta)^H\mathbf{E}_n\mathbf{E}_n^H\mathbf{a}(\theta) \tag{26}$$

Note that the squared norm is taken because $\mathbf{E}_n^H\mathbf{a}(\theta)$ results in a complex vector; by squaring it, a real scalar value representing the magnitude of the projection is obtained.

If θ is a target, $d^2(\theta) \approx 0$. Since peaks are preferred over nulls, the MUSIC pseudo-spectrum is defined as the inverse:

$$P_{MUSIC}(\theta) = \frac{1}{\mathbf{a}(\theta)^H\mathbf{E}_n\mathbf{E}_n^H\mathbf{a}(\theta)} \tag{27}$$

The peaks of $P_{MUSIC}(\theta)$ correspond to the estimated directions of arrival.

3.3.5 Pseudocode for the algorithm

```
Algorithm 2 MUSIC Algorithm
Require: Array data matrix $X \in \mathbb{C}^{M\times N}$, Number of sources $K$, Sensor spacing $d$, wavelength $\lambda$, Angular scan grid $\Theta = \{\theta_i\}_{i=1}^{I}$
Ensure: MUSIC pseudospectrum $P_{\text{MUSIC}}(\theta)$ over $\Theta$
Estimate covariance matrix:
    $R_x \leftarrow \frac{1}{N} X X^H$
Eigen-decomposition:
    $R_x = E D E^H$
Sort eigenvalues in descending order
Noise subspace:
    $E_n \leftarrow E(:, K+1 : M)$
for $i = 1$ to $I$ do
    Steering vector:
        $a(\theta_i) = \exp\left(-j\frac{2\pi}{\lambda} d\, m\, \sin\theta_i\right)$
    MUSIC spectrum:
        $P_{\text{MUSIC}}(\theta_i) = \frac{1}{\|E_n^H a(\theta_i)\|_2^2}$
end for
Normalize the pseudospectrum to its maximum value
Identify the $K$ largest peaks in $P_{\text{MUSIC}}(\theta)$ as DOA estimates
```

3.4 Forward-Backward Spatial Smoothing

Standard subspace algorithms (like MUSIC) rely on the source covariance matrix $\mathbf{R}_s$ being "full rank." This condition is met when sources are uncorrelated. However, it fails in environments with coherent (perfectly correlated) signals, such as multipath propagation. In these cases, the rank of $\mathbf{R}_s$ is less than K. To solve this problem, the Forward-Backward Spatial Smoothing (FBSS) algorithm is implemented.

3.4.1 Mathematical Model of Signal Coherence

Consider a scenario with K coherent sources. Here, the k-th source is modeled as a copy of the first source, but with a different scale and phase. The source signal vector $\mathbf{s}(t)$ is expressed as:

$$\mathbf{s}(t) = \begin{bmatrix} s_1(t) \\ \beta_2 s_1(t) \\ \vdots \\ \beta_K s_1(t) \end{bmatrix} = s_1(t)\mathbf{c} \tag{28}$$

where $\mathbf{c} = [1, \beta_2, \ldots, \beta_K]^T$ is the coherence vector. The resulting source covariance matrix is:

$$\mathbf{R}_s = E[\mathbf{s}(t)\mathbf{s}^H(t)] = E[|s_1(t)|^2]\mathbf{c}\mathbf{c}^H \tag{29}$$

Because $\mathbf{c}\mathbf{c}^H$ is created from a single vector, its rank is exactly 1. As a result, the signal subspace collapses into a single dimension, regardless of the number of sources K. This makes standard MUSIC ineffective.

3.4.2 Subarray Formulation and Forward Smoothing

The Uniform Linear Array (ULA) with M sensors is divided into L overlapping subarrays. Each subarray contains m sensors, such that $M = L + m - 1$.

Let $\mathbf{x}_l(t)$ be the data vector received by the l-th subarray (sensors l to $l+m-1$). The steering matrix for this subarray is defined using the first m phase shifts. This reference matrix, $\mathbf{A}_m \in \mathbb{C}^{m\times K}$, is defined as:

$$\mathbf{A}_m = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e^{-j\phi_1} & e^{-j\phi_2} & \cdots & e^{-j\phi_K} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j(m-1)\phi_1} & e^{-j(m-1)\phi_2} & \cdots & e^{-j(m-1)\phi_K} \end{bmatrix} \tag{30}$$

where $\phi_k = \frac{2\pi}{\lambda} d \sin\theta_k$ is the phase shift between sensors for the k-th source.

Because the array structure is uniform , the signal at the l-th subarray is simply a phase-shifted version of the first. A diagonal phase matrix $\mathbf{\Phi}$ is introduced:

$$\mathbf{\Phi} = \mathrm{diag}\left(e^{-j\phi_1}, e^{-j\phi_2}, \ldots, e^{-j\phi_K}\right) \tag{31}$$

The output of the l-th subarray is expressed as:

$$\mathbf{x}_l(t) = \mathbf{A}_m \mathbf{\Phi}^{l-1}\mathbf{s}(t) + \mathbf{n}_l(t) \tag{32}$$

The covariance matrix for the l-th subarray, $\mathbf{R}_l$, is derived as:

$$\mathbf{R}_l = E[\mathbf{x}_l\mathbf{x}_l^H] = \mathbf{A}_m\mathbf{\Phi}^{l-1}\mathbf{R}_s(\mathbf{\Phi}^{l-1})^H\mathbf{A}_m^H + \sigma^2\mathbf{I} \tag{33}$$

The Forward Spatial Smoothing technique averages these subarray covariances:

$$\mathbf{R}_{fwd} = \frac{1}{L}\sum_{l=1}^{L}\mathbf{R}_l = \mathbf{A}_m\left[\frac{1}{L}\sum_{l=1}^{L}\mathbf{\Phi}^{l-1}\mathbf{R}_s(\mathbf{\Phi}^{l-1})^H\right]\mathbf{A}_m^H + \sigma^2\mathbf{I} \tag{34}$$

The term in the brackets is the *smoothed source covariance matrix*, $\bar{\mathbf{R}}_s$. Because the phase shifts in $\mathbf{\Phi}$ are different for each angle, the summation restores the rank of $\bar{\mathbf{R}}_s$ to K (provided $L \geq K$).

3.4.3 Backward Smoothing and Final Algorithm

To improve stability and use the full capacity of the array, Backward Smoothing is also used. This method uses the complex conjugate of the data in reverse order. An Exchange Matrix $\mathbf{J}_m$ (size $m \times m$) is defined as:

$$\mathbf{J}_m = \begin{bmatrix} 0 & \cdots & 1 \\ \vdots & & \vdots \\ 1 & \cdots & 0 \end{bmatrix} \tag{35}$$

The backward covariance matrix $\mathbf{R}_{bwd}$ is calculated as:

$$\mathbf{R}_{bwd} = \mathbf{J}_m \mathbf{R}^*_{fwd} \mathbf{J}_m \tag{36}$$

This step simulates a virtual array looking from the opposite direction. The final Forward-Backward Smoothed covariance matrix is obtained by:

$$\mathbf{R}_{FB} = \frac{1}{2}(\mathbf{R}_{fwd} + \mathbf{R}_{bwd}) \tag{37}$$

Finally, the standard eigen-decomposition is applied to $\mathbf{R}_{FB}$ to find the noise subspace $\mathbf{E}_n$. The MUSIC spectrum is computed using the subarray steering vector $\mathbf{a}_m(\theta)$:

$$P_{FB}(\theta) = \frac{1}{\mathbf{a}_m^H(\theta)\mathbf{E}_n\mathbf{E}_n^H\mathbf{a}_m(\theta)} \tag{38}$$

This ensures that the orthogonality condition $\mathbf{a}_m^H(\theta)\mathbf{E}_n \approx 0$ holds true, even for perfectly correlated sources.

3.4.4 Pseudocode for the algorithm

Algorithm 3 Forward–Backward Spatial Smoothing (FB Averaging)

```
Require: Array data matrix $X \in \mathbb{C}^{M \times N}$, Number of sources $K$, Subarray size $m$
Ensure: Forward–Backward smoothed covariance matrix $R_{\text{FB}}$
Number of overlapping subarrays: $L \leftarrow M - m + 1$
Initialize forward covariance: $R_{\text{fwd}} \leftarrow 0_{m \times m}$
for $\ell = 1$ to $L$ do
    Extract subarray data: $X_\ell \leftarrow X(\ell : \ell + m - 1, :)$
    Subarray covariance: $R_\ell \leftarrow \frac{1}{N} X_\ell X_\ell^H$
    Accumulate: $R_{\text{fwd}} \leftarrow R_{\text{fwd}} + R_\ell$
end for
Average: $R_{\text{fwd}} \leftarrow \frac{1}{L} R_{\text{fwd}}$
Form exchange matrix $J_m$ (ones on anti-diagonal)
Backward covariance:
    $R_{\text{bwd}} \leftarrow J_m R^*_{\text{fwd}} J_m$
Forward–Backward average:
    $R_{\text{FB}} \leftarrow \frac{1}{2}(R_{\text{fwd}} + R_{\text{bwd}})$
Apply MUSIC to $R_{\text{FB}}$ to estimate DOAs
```

4 Implementation and Simulation Results

4.1 Implementation Details

All simulations were implemented in MATLAB for a uniform linear array (ULA). The carrier frequency was set to $f_c = 2.4$ GHz. Half-wavelength spacing was used, so $d = \lambda/2$.

Initially, $M = 16$ antennas, $K = 2$ sources, $N = 1000$ snapshots, and SNR $= 10$ dB were selected. The spectra is plotted in dB.

To study performance, one parameter was changed at a time. The effect of antenna number was tested with $M = 4$ and $M = 16$. The effect of SNR was tested with -20 dB and 20 dB. The effect of snapshots was tested with $N = 3$ and $N = 1000$.

For coherent sources, the second signal is generated as a scaled and phase-shifted copy of the first. Forward–backward spatial smoothing is then applied with overlapping subarrays ($m = 7$), and MUSIC is run on the smoothed covariance matrix.

4.2 Simulation Results

4.2.1 Simulation 1: The Most Basic Scenario

The first simulation is set up to test the algorithms for the most basic scenario which is the presence of a single source. Fig.3 confirms that both algorithms identified the target correctly. Conventional Beamforming method shows a wide main lobe while MUSIC produces a sharp peak which provides a more precise result.

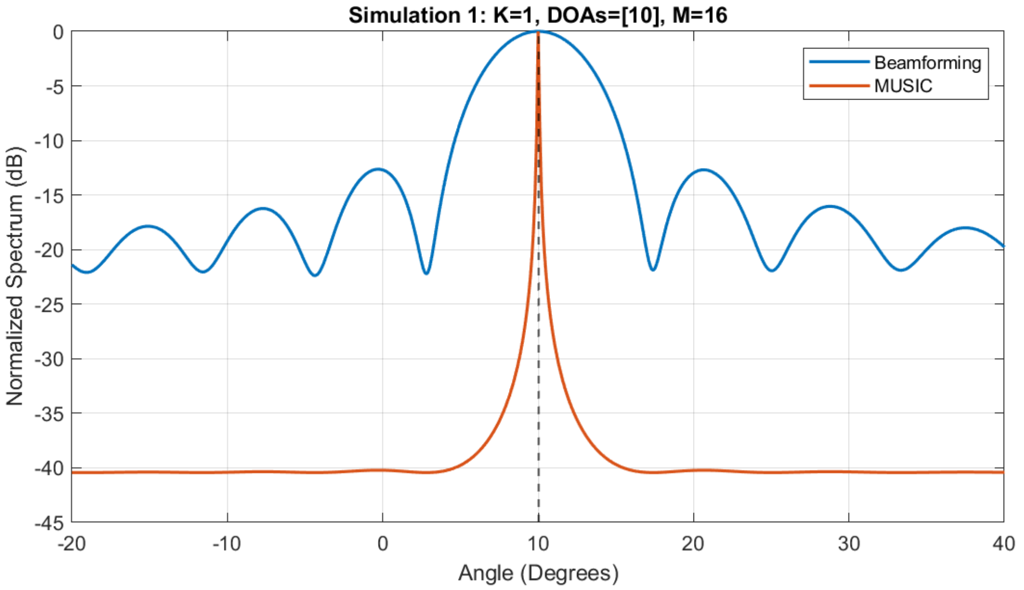

Figure 3: Single source case

4.2.2 Simulation 2: Two Sources

Fig. 4 shows the situation of the existence of 2 sources. Both algorithms provides clear information about the DOAs of the sources. Still, MUSIC exhibits a significantly lower noise floor compared to the high sidelobes of the Beamformer.

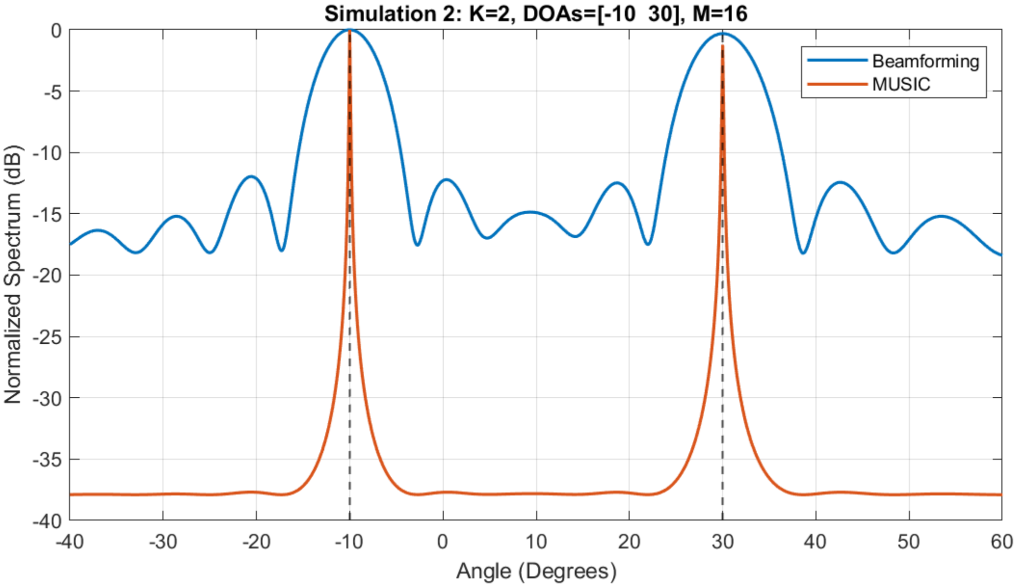

Figure 4: Two sources case

4.2.3 Simulation 3: Disadvantage of Beamforming

Fig. 5 highlights the critical difference between the methods. The 5° separation is below the array's beamwidth which causes Beamforming method to fail to differentiate two seperate sources. On the other hand, MUSIC successfully identifies the two seperate sources proving that it overcomes the Rayleigh limit.

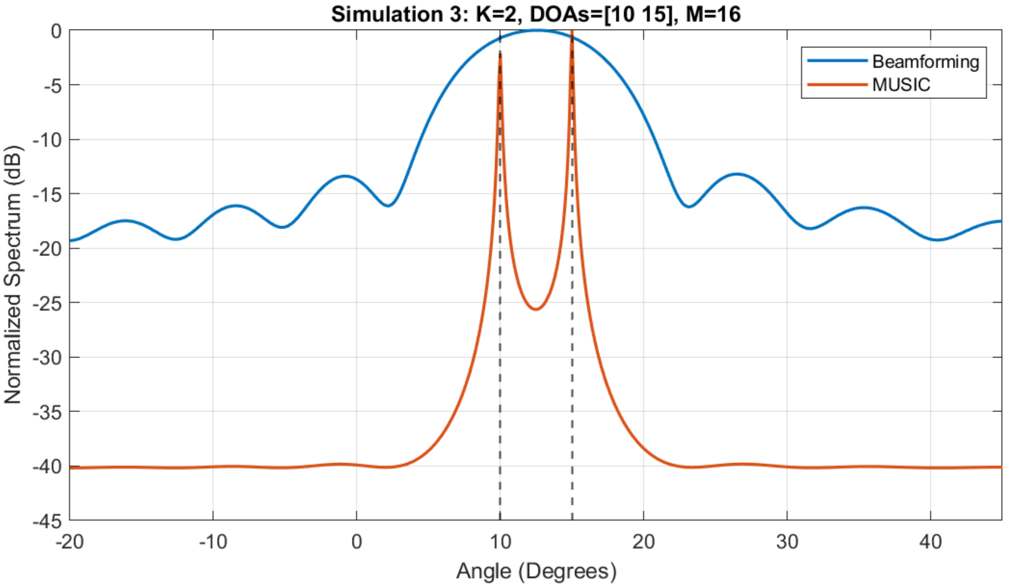

Figure 5: Small seperation case

4.2.4 Simulation 4: Effect of Antenna Number

Fig. 6 demonstrates the relationship between the number of array elements (M) and angular resolution. With a low number of antennas ($M = 4$), the array aperture is not enough which results in a wide beamwidth that causes the Beamforming algorithm to fail to identify two seperate sources and including them both in a single peak. On the other hand, increasing the number of elements to $M = 16$ narrows the beamwidth and sharpens the MUSIC spectrum which allows the clear identification of both sources.

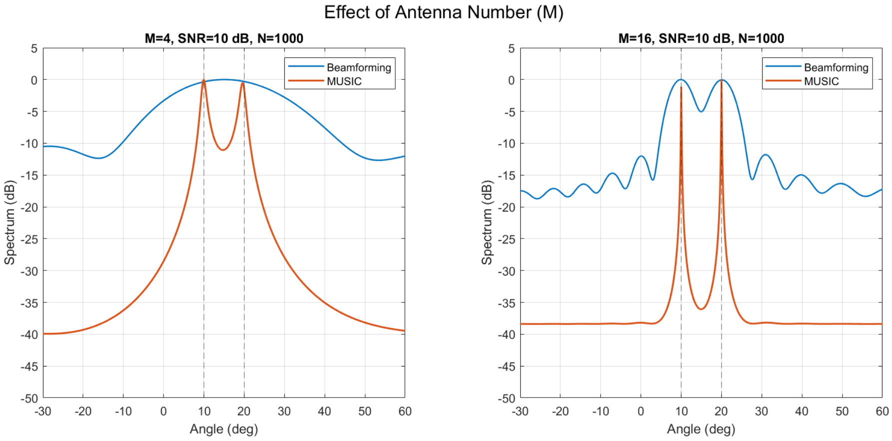

Figure 6: Analysis of Antenna Number

4.2.5 Simulation 5: Effect of Signal-to-Noise Ratio

Fig. 7 shows the effect of Signal-to-Noise Ratio on the performance of both Beamforming and MUSIC. At a very low SNR of -20 dB, the received data are dominated by noise, which results in a poorly estimated covariance matrix. As a consequence, Beamforming produces broad and ambiguous responses, while MUSIC fails to form distinct peaks due to the loss of clear subspace separation. When the SNR is increased to 20 dB, the signal components become dominant. In this case, Beamforming shows clearer main lobes, and MUSIC produces sharp and well-defined peaks at the correct directions of arrival.

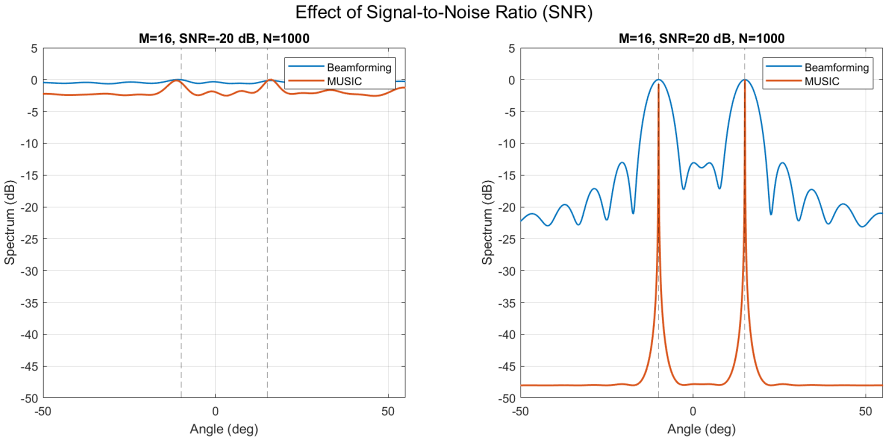

Figure 7: Analysis of SNR

4.2.6 Simulation 6: Effect of Number of Snapshots

Fig. 8 shows the effect of the number of snapshots (N) on the performance of both Beamforming and MUSIC. When only a small number of snapshots ($N = 3$) is used, the estimated covariance matrix is poorly conditioned. As a result, Beamforming exhibits irregular and distorted responses, while MUSIC produces unstable spectra with false or merged peaks. When the number of snapshots is increased to $N = 1000$, the covariance estimate becomes more reliable. In this case, although Beamforming still can't identify the two seperate targets, MUSIC provides sharp and stable peaks at the correct directions of arrival.

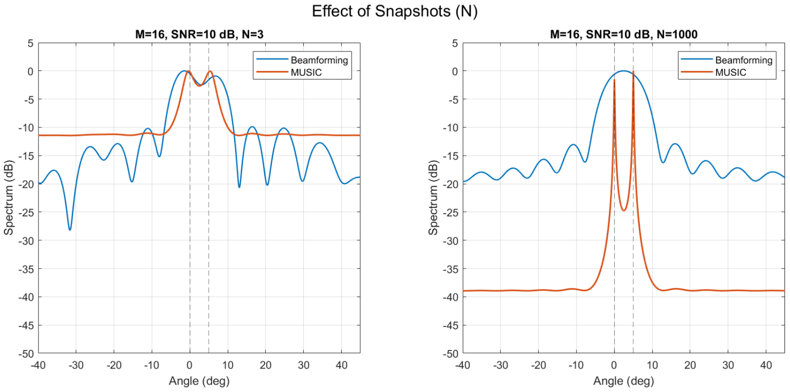

Figure 8: Analysis of Number of Snapshots

4.2.7 Simulation 7: Effect of Spatial Smoothing on Coherent Signals

Fig. 9 demonstrates the limitation of the standard MUSIC algorithm when it is used with coherent sources such as multipath signals. As shown in the top plot, standard MUSIC fails to find the correct angles because the signal covariance matrix loses rank, which causes the signal and noise subspaces to merge. The bottom plot shows that applying Forward-Backward Spatial Smoothing restores the rank of the covariance matrix which allows the algorithm to correctly detect the DOA peaks.

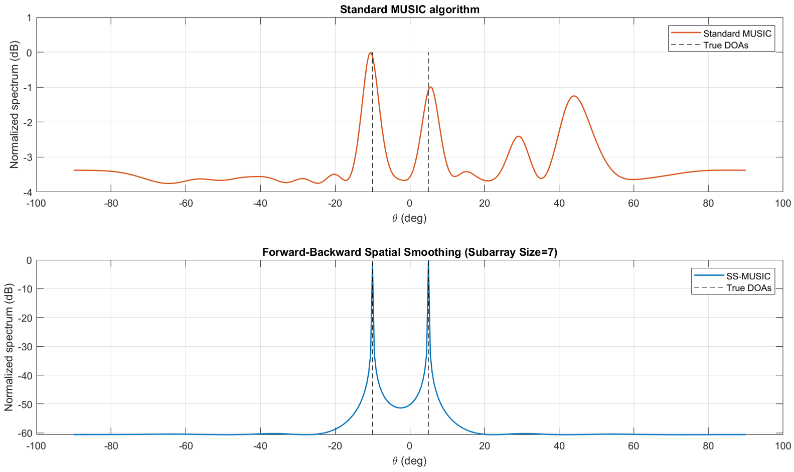

Figure 9: Coherent Sources Case

5 Conclusion

This project presented a comprehensive comparative analysis of array-based Direction of Arrival (DOA) estimation techniques, specifically focusing on Conventional Beamforming (Delay-and-Sum) and the subspace-based MUSIC algorithm. Through a series of MATLAB simulations using a Uniform Linear Array (ULA), the performance of these algorithms was evaluated under varying environmental conditions and physical array configurations.

The simulation results clearly show that the MUSIC algorithm performs better than Conventional Beamforming, especially in terms of spatial resolution and resistance to noise. While both methods successfully identified sources in basic scenarios with wide angular separation, the Conventional Beamformer exhibited significant performance degradation when the angular separation between sources fell below the Rayleigh resolution limit. In contrast, the MUSIC algorithm used the orthogonality between the signal and noise subspaces to correctly separate these closely spaced sources, effectively overcoming the physical limits of the array aperture.

Furthermore, the analysis highlighted the sensitivity of these algorithms to system parameters. It was observed that increasing the number of antenna elements (M) narrows the beamwidth and enhances the sharpness of the estimated peaks. Conversely, the performance of the MUSIC algorithm was shown to be heavily dependent on the accuracy of the covariance matrix estimation. Simulations with a low Signal-to-Noise Ratio (SNR of 20 dB) and a small number of snapshots ($N = 3$) caused the algorithm to fail, leading to false peaks and numerical instability. This confirms that while MUSIC offers high resolution, it requires a statistically stable covariance estimate, achieved through sufficient sampling and adequate signal power.

Finally, the project addressed the critical limitation of the standard MUSIC algorithm in the presence of coherent signals, such as those caused by multipath propagation. The standard algorithm failed to detect coherent sources due to the rank deficiency of the source covariance matrix, which causes the signal and noise subspaces to merge. The implementation of Forward-Backward Spatial Smoothing (FBSS) proved to be an effective solution. By averaging the covariance matrices of overlapping subarrays, FBSS successfully restored the matrix rank, allowing for the accurate estimation of DOAs for perfectly correlated signals.

In summary, while Conventional Beamforming remains a computationally simple and robust approach for single-source detection, the MUSIC algorithm is the better choice for high-precision modern radar and communication systems requiring the separation of multiple closely spaced targets. For practical implementations in multipath environments, implementing MUSIC with spatial smoothing techniques is essential to ensure reliability.

References

[1] Smith, M. S. (1989, September). Phased array fundamentals. In IEE Tutorial Meeting on Phased Array Radar (pp. 1-1). IET.

[2] Steyskal, H. (1988). Digital beamforming. In *1988 18th European Microwave Conference* (pp. 49–57). IEEE.

[3] Van Veen, B. D., & Buckley, K. M. (1988). Beamforming: A versatile approach to spatial filtering. *IEEE ASSP Magazine*, *5*(2), 4–24.

[4] Khan, Z. I., Kamal, M. M., Hamzah, N., Othman, K., & Khan, N. I. (2008). Analysis of performance for multiple signal classification (MUSIC) in estimating direction of arrival. In *2008 IEEE International RF and Microwave Conference* (pp. 524–529). IEEE.

[5] Thompson, J. S., Grant, P. M., & Mulgrew, B. (2002). Performance of spatial smoothing algorithms for correlated sources. IEEE Transactions on Signal Processing, 44(4), 1040-1046.

[6] Rao, C., Ding, Z., Dobre, O. A., & Dai, X. (2024). A general analytical framework for the resolution of near-field beamforming. IEEE Communications Letters, 28(5), 1171-1175.

[7] Salama, A. A. (2025). Direction of Arrival Estimation: A Tutorial Survey of Classical and Modern Methods. arXiv preprint arXiv:2508.11675.

[8] Gamba, J. (2019). Radar signal processing for autonomous driving. Springer. https://doi.org/10.1007/978-981-13-9193-4